

Do contrastive embeddings converge to a log-probability? InfoNCE vs. LeCun’s MSE + SIGReg (LeJEPA)

1 Question

Given a contrastive-style encoder trained to its optimum, does the distance between two (class) centers in embedding space converge to a log-probability? We compare two regimes: (i) InfoNCE / SimCLR-style losses, and (ii) LeCun’s MSE + SIGReg objective (LeJEPA, arXiv:2511.08544).

2 InfoNCE: the log-density-ratio comes from the partition function

For anchor x , positive x^+ , temperature τ , and critic f ,

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{f(x, x^+)}{\tau} + \log \sum_k \exp \frac{f(x, x_k)}{\tau}. \quad (1)$$

With unbounded negatives and sufficient capacity, the minimizing critic is the pointwise mutual information (Oord et al.; Wang & Isola):

$$\frac{f^*(x, x^+)}{\tau} = \log \frac{p(x^+ | x)}{p(x^+)} + \text{const} = \text{PMI}(x, x^+). \quad (2)$$

The *softmax denominator* is what forces the critic onto a log-ratio: InfoNCE is implicitly a density-ratio estimator. On the unit sphere the *similarity* is what lands on a log-probability, and distance is affine in it:

$$d^2(z_i, z_k) = 2 - 2 \cos \theta_{ik} = 2 - 2\tau \text{ PMI}(x_i, x_k) / \text{scale}. \quad (3)$$

Supervised case. With supervised positives (same-class x^+), $p(x^+ | x)$ is the same-class conditional, and the center-to-center object becomes a *log-likelihood-ratio* between class conditionals $p(\cdot | a)$ and $p(\cdot | b)$ — a measure of class confusability/overlap.

Two caveats.

1. **Non-collapse required.** If the embedding dimension is too small (and τ small), the supervised optimum is a simplex equiangular tight frame (neural collapse): all class centers equidistant, and the likelihood information is destroyed.
2. **Affine, and a ratio.** The readout is affine in a log-density *ratio*, with a τ -dependent scale and an unidentified additive constant — not a calibrated log-likelihood.

3 LeJEPa (MSE + SIGReg): the mechanism is removed by design

The objective has no partition function, no negatives, no critic:

$$\mathcal{L} = \underbrace{\|z_{\text{pred}} - z_{\text{target}}\|^2}_{\text{alignment}} + \lambda \underbrace{\mathcal{L}_{\text{SIGReg}}}_{\text{marginal} \rightarrow \mathcal{N}(0, I)}, \quad \mathcal{L}_{\text{SIGReg}} = \sum_k |\phi_{\text{emp}}(\omega_k) - \phi_{\mathcal{N}}(\omega_k)|^2, \quad (4)$$

where SIGReg is an Epps–Pulley characteristic-function test pinning the *aggregate* embedding marginal to an isotropic Gaussian. Because there is no softmax denominator, *the InfoNCE→PMI argument does not apply*: nothing drives the pairwise distance to a log-likelihood-ratio.

What is a log-probability here. The constraint makes the *norm*, not the pairwise distance, affine in a log-density — and it is the density under the *imposed prior*, not the data:

$$\log p_{\text{ref}}(z) = -\frac{1}{2}\|z\|^2 + \text{const} \implies \|z\|^2 = -2 \log p_{\text{ref}}(z). \quad (5)$$

Recovering the true *data* log-likelihood would additionally require the encoder’s change-of-variables Jacobian (as in a normalizing flow); SIGReg constrains only the marginal, not the Jacobian, so data log-likelihood is *not* recovered.

Supervised / class centers. Supervised MSE (pull same-class together) + SIGReg (spread the set to $\mathcal{N}(0, I)$) yields Gaussian class blobs packed so the mixture $\approx \mathcal{N}(0, I)$. Then the class-center distance is a Mahalanobis/Gaussian separation, which equals a log-likelihood-ratio *only under an added LDA-style assumption*: each class conditional Gaussian with shared covariance. SIGReg guarantees only that the *aggregate* is Gaussian, not the per-class conditionals. With balanced classes the centers pack near-symmetrically, so distances reflect class count/priors (packing geometry), not confusability — the Gaussian-packing analogue of the ETF collapse.

4 Summary

	InfoNCE / SimCLR	LeJEPa (MSE + SIGReg)
Similarity →	$\log \frac{p(x^+ x)}{p(x^+)}$ (PMI)	nothing (no critic)
Pinned to a log-prob	pairwise distance (affine in $-\log$ -ratio)	the <i>norm</i> : $\ z\ ^2 = -2 \log p_{\text{ref}}$
Center dist. = log-lik. ratio?	yes, up to affine + non-collapse	only under equal-cov. Gaussian assumption

Bottom line. InfoNCE’s distance encodes an *estimated* log-density-ratio via its partition function. LeCum’s MSE + SIGReg deliberately replaces that with an *imposed* marginal prior: you gain provable distributional control and lose the automatic “distance = log-likelihood” readout. To make class-center distances into log-likelihood-ratios under SIGReg, you must add a per-class second-moment constraint; the aggregate Gaussian alone will not do it.

5 An NPLM-based contrastive loss

The quantity we want — a *calibrated* log-likelihood-ratio — is exactly what the New Physics Learning Machine (NPLM, D’Agnolo & Wulzer) estimates by construction, so it is a natural loss to build a contrastive objective on.

Reframing. Both SimCLR and the supervised question are asking for the pointwise mutual information

$$\text{PMI}(x, x') = \log \frac{p(x, x')}{p(x)p(x')}, \quad (6)$$

where the joint $p(x, x')$ is the distribution of *positive pairs* (augmentations of the same point, or same-class pairs) and the reference $p(x)p(x')$ is *independent pairs* (a shuffled batch). InfoNCE estimates this ratio with a coupled softmax; NPLM estimates the same ratio with a maximum-likelihood (Neyman–Pearson) loss.

The loss. NPLM parametrizes the alternative density as $n(x) = n_{\text{ref}}(x) e^{f(x)}$ and minimizes $\mathcal{L}[f] = \sum_{\text{ref}}(e^f - 1) - \sum_{\text{data}} f$. Ported to pairs with a critic $g(x, x')$,

$$\mathcal{L}[g] = \mathbb{E}_{(x, x') \sim p(x)p(x')} [e^{g(x, x')} - 1] - \mathbb{E}_{(x, x') \sim p(x, x')} [g(x, x')]. \quad (7)$$

The stationarity condition $p(x)p(x') e^{g^*} = p(x, x')$ gives

$$g^*(x, x') = \log \frac{p(x, x')}{p(x)p(x')} = \text{PMI}(x, x'). \quad (8)$$

Crucially the $e^g - 1$ term forces $\mathbb{E}_{\text{ref}}[e^{g^*}] = 1$, which pins the additive constant: unlike InfoNCE, the recovered value is the *absolute*, calibrated log-likelihood-ratio.

Distance geometry. Choosing

$$g(x, x') = -\frac{1}{2} \|z(x) - z(x')\|^2 + b(x) + b(x'), \quad (9)$$

with a learned log-density bias b , the optimum yields

$$\|z(x) - z(x')\|^2 = -2 \text{PMI}(x, x') + 2b(x) + 2b(x'), \quad (10)$$

so squared distance is exactly affine in the negative *calibrated* log-likelihood-ratio, with b absorbing the per-point marginal density.

Test statistic. $t = -2 \mathcal{L}_{\min}$ is an asymptotic log-likelihood-ratio test (Wilks) — the same object trained is the signal-detection statistic, which is the point of NPLM.

Hybrid with LeJEPa. Keep SIGReg to pin the marginal $p(z) = \mathcal{N}(0, I)$ (so $\|z\|^2 = -2 \log p_{\text{ref}}$) and add the NPLM ratio head (7) for calibrated pairwise log-likelihood-ratios. This gives both a controlled marginal geometry and pairwise distances that are genuine calibrated log-likelihood-ratios — the “distance = log-probability” property neither loss delivers alone.

5.1 Pros and cons

Pros.

- **Calibration.** The $e^g - 1$ term fixes the additive constant, so the critic recovers the *absolute* log-likelihood-ratio, not “PMI+const” as in InfoNCE. This is the “distance = log-likelihood” readout, genuinely calibrated.

- **No coupled partition function.** The reference expectation is separable; there is no softmax denominator locking many negatives together. One needs enough shuffled pairs, not a large coupled negative set.
- **Built-in test statistic.** $t = -2\mathcal{L}_{\min}$ is an asymptotic log-likelihood-ratio test, directly usable for signal detection — the trained object and the test statistic coincide.
- **Weights and imbalance.** Class weights / imbalance enter as simple sample weights, without reweighting a normalization constant.
- **Interpretable geometry.** With the distance parametrization, pairwise distances and class-center separations are calibrated log-likelihood-ratios of confusability, free of the ETF-collapse scale ambiguity.

Cons.

- **Exponential instability.** e^g diverges when g is large-positive on reference pairs — the classic NPLM training pathology. Requires weight clipping / bounding $\|g\|$, a kernel version (Falkon-NPLM), or regularization to control the variance of $\mathbb{E}_{\text{ref}}[e^g]$.
- **Reference-sample cost.** It remains density-ratio estimation; the calibration is only as good as the shuffled-pair coverage of $p(x)p(x')$.
- **Identifiability of the distance form.** The bias b must be expressive enough to represent marginal-density variation, or the distance term is forced to explain structure it cannot represent.
- **No automatic marginal control.** Unlike SIGReg, the loss says nothing about the shape of $p(z)$; if a controlled marginal is wanted it must be added (hence the hybrid above).
- **Higher-variance gradients.** The exponential term generally yields noisier gradients than the log-softmax of InfoNCE, so it can need smaller steps / more careful optimization.

6 Building an embedded log-likelihood space (NPLM + SIGReg)

The aim of this section is a concrete objective, structured like InfoNCE (an encoder $z = h_\theta(x)$, positive pairs, in-batch negatives), whose optimum makes the *embedding geometry itself* the log-likelihood surface — not up to an unidentified constant, but “in the true sense.”

6.1 Design target: what “true log-likelihood space” means

We ask the encoder $z = h_\theta(x) \in \mathbb{R}^d$ to satisfy two readouts simultaneously, against a fixed reference $q(z) = \mathcal{N}(0, I)$:

$$\text{(marginal)} \quad \log p(x) = -\frac{1}{2}\|z\|^2 + c_0, \tag{11}$$

$$\text{(interaction)} \quad \log \frac{p(x, x')}{p(x)p(x')} = \langle z, z' \rangle. \tag{12}$$

Adding (11) for x and x' to (12) gives the *joint* log-density as a single quadratic,

$$\log p(x, x') = -\frac{1}{2}\|z\|^2 - \frac{1}{2}\|z'\|^2 + \langle z, z' \rangle + c_1 = -\frac{1}{2}\|z - z'\|^2 + c_1. \tag{13}$$

So the design target is precisely: *marginals standard Gaussian, positive pairs Gaussian in the embedding difference*. Every likelihood object — marginal, conditional, ratio, joint — is then a quadratic in z , which is what we mean by the embedding *being* log-likelihood space. Equations (11) and (12) are supplied by two different terms: SIGReg delivers the marginal, NPLM delivers the interaction.

6.2 The embedded NPLM critic and its minimizer

Take the InfoNCE-style critic — inner product plus a learned per-point bias —

$$g_\theta(x, x') = \langle z, z' \rangle + b_\theta(x) + b_\theta(x'), \quad z = h_\theta(x), \quad (14)$$

and plug it into the NPLM density-ratio loss between the joint (positives) and the product of marginals (independent pairs),

$$\mathcal{L}_{\text{NPLM}}[g] = \mathbb{E}_{p(x)p(x')} [e^g - 1] - \mathbb{E}_{p(x, x')} [g]. \quad (15)$$

Its unconstrained functional derivative, $\delta \mathcal{L} / \delta g = p(x)p(x')e^g - p(x, x')$, vanishes at

$$g^*(x, x') = \log \frac{p(x, x')}{p(x)p(x')} = \text{PMI}(x, x'). \quad (16)$$

For the *constrained* family (14) the minimizer is a projection: using $p(x, x') = p(x)p(x')e^{g^*}$,

$$\mathcal{L}_{\text{NPLM}}[g] - \mathcal{L}_{\text{NPLM}}[g^*] = \mathbb{E}_{p(x)p(x')} [e^g - e^{g^*} - e^{g^*}(g - g^*)] = D_\phi(g^* \| g) \geq 0, \quad (17)$$

the Bregman divergence generated by the convex $\phi(t) = e^t$ under the reference measure. Hence the encoder learns the *best exponential-Bregman approximation* of PMI representable by (14); if PMI is genuinely bilinear-plus-separable, the fit is exact and $\langle z, z' \rangle + b(x) + b(x') = \text{PMI}(x, x')$.

6.3 The residual gauge, and how SIGReg removes it

Equation (14) has exactly the gauge freedom that leaves InfoNCE’s constant unidentified:

$$z \mapsto Rz + a, \quad b(x) \mapsto b(x) - \langle a, z \rangle - \frac{1}{2}\|a\|^2 + \beta, \quad (18)$$

for any rotation R , shift a , constant β — all leave g (hence $\mathcal{L}_{\text{NPLM}}$) invariant. So NPLM alone fixes the *interaction* (12) but not the split between z and b : the map to “log-likelihood space” is undetermined. Impose SIGReg on the marginal,

$$\mathcal{L}_{\text{SIGReg}} = \sum_k |\phi_{\text{emp}}(\omega_k) - \phi_{\mathcal{N}}(\omega_k)|^2 \xrightarrow{\min} p(z) = \mathcal{N}(0, I). \quad (19)$$

This kills the shift/rotation freedom (only R with $a = 0$ survives, an irrelevant frame choice) and, via (11), *identifies the bias as the marginal log-density*,

$$b_\theta(x) \longrightarrow -\frac{1}{2}\|z\|^2 + \text{const}. \quad (20)$$

Substituting back into (14) collapses the critic to the pure distance form $g = -\frac{1}{2}\|z - z'\|^2 + \text{const}$, and (13) holds. The two terms are therefore complementary by design: **NPLM pins the interaction and its absolute scale** ($\mathbb{E}_{\text{ref}}[e^{g^*}] = 1$), **SIGReg pins the marginal and removes the residual gauge**. Neither alone yields (11)–(13); together they do.

6.4 The combined objective and its minibatch estimator

$$\mathcal{L} = \underbrace{\mathbb{E}_{p(x)p(x')} [e^{g_\theta} - 1] - \mathbb{E}_{p(x,x')} [g_\theta]}_{\text{NPLM interaction}} + \lambda \underbrace{\sum_k |\phi_{\text{emp}}(\omega_k) - \phi_{\mathcal{N}}(\omega_k)|^2}_{\text{SIGReg marginal}} \quad (21)$$

For a batch $\{(x_i, x_i^+)\}_{i=1}^N$ with $g_{ij} = g_\theta(x_i, x_j^+)$, use the diagonal as positives and the off-diagonal (shuffled) pairs as the product-of-marginals reference:

$$\hat{\mathcal{L}}_{\text{NPLM}} = \frac{1}{N(N-1)} \sum_{i \neq j} (e^{g_{ij}} - 1) - \frac{1}{N} \sum_i g_{ii}. \quad (22)$$

This reuses *exactly* the InfoNCE batch geometry; the only change is the loss functional:

$$\hat{\mathcal{L}}_{\text{InfoNCE}} = -\frac{1}{N} \sum_i \log \frac{e^{g_{ii}}}{\sum_j e^{g_{ij}}} \quad \text{vs.} \quad \hat{\mathcal{L}}_{\text{NPLM}} = \frac{1}{N(N-1)} \sum_{i \neq j} e^{g_{ij}} - \frac{1}{N} \sum_i g_{ii} - 1. \quad (23)$$

InfoNCE is the *self-normalized* (per-anchor softmax) estimator of the same log-ratio; NPLM is the *un-normalized maximum-likelihood* estimator, which is why it retains the absolute normalization that InfoNCE discards. Reading g_{ii} off the diagonal as $-\frac{1}{2}\|z_i - z_i^+\|^2$ (once SIGReg is active) makes the positive term a plain squared-distance pull, and the reference term a repulsion weighted by $e^{-\frac{1}{2}\|z_i - z_j^+\|^2}$.

6.5 Result

At the joint optimum the embedding satisfies the three quadratic identities

$$\log p(x) = -\frac{1}{2}\|z\|^2 + c_0, \quad \log \frac{p(x, x')}{p(x)p(x')} = \langle z, z' \rangle, \quad \log p(x, x') = -\frac{1}{2}\|z - z'\|^2 + c_1, \quad (24)$$

so marginal densities, pairwise log-likelihood-ratios, and joint log-densities are all read directly off the geometry — an embedding that approximates log-likelihood space in the true (calibrated, gauge-fixed) sense. The approximation is exact when PMI lies in the bilinear-plus-separable family (14); otherwise (17) says the encoder returns the closest such surface in exponential-Bregman divergence. The practical caveats of §5.1 (exponential instability of e^g , reference-pair coverage) carry over and set the sole implementation burden.

7 The exp-34e hybrid, and a configurable NPLM geometry

The SuperSig codebase already contains a hybrid of the contrastive and SIGReg families (`experiments/34e_hybrid` via `train_simclr_sigreg`). This section places it in the framework above, identifies why it stops short of the §6 identities, and shows that a single loss class — with the NPLM interaction as one option — spans it and many neighbours.

7.1 What exp-34e does, in this language

For two augmented views v_1, v_2 with raw embeddings $z = h_\theta(v)$, the hybrid half optimizes

$$\mathcal{L}_{34e} = \underbrace{\text{NTXent}(\hat{z}, \hat{z}')}_{\text{instance positives, } \hat{z}=z/\|z\|} + \lambda \underbrace{\text{SIGReg}(z)}_{\text{raw } z \rightarrow \mathcal{N}(0, I)}, \quad \hat{z} = z/\|z\|. \quad (25)$$

In the vocabulary above: NT-Xent is the *self-normalized* (softmax) estimator of PMI (§5, §6.4) with *instance* positives, and SIGReg supplies the marginal (§3). The supervised twin `train_supcon_sigreg` just swaps instance positives for same-label positives. So exp-34e is already the “InfoNCE + SIGReg” hybrid of §5 — with one consequential detail.

7.2 Why it stops short: two geometries

NT-Xent acts on the *normalized* embedding $\hat{z}$ (angles on the unit sphere; scale-invariant), whereas SIGReg acts on the *raw* embedding z (it is entirely about the radial and coordinate distribution). The interaction term is blind to $\|z\|$; the marginal term is largely about $\|z\|$. The two therefore speak different geometries and are only loosely coupled: one obtains features whose *angles* encode PMI+const sitting inside a raw distribution nudged toward Gaussian — not the single coherent surface of (13). This is precisely why exp-34e yields good, calibrated-looking features but not the log-likelihood identities of §6: the softmax discards the constant *and* the interaction lives in a different space than the marginal constraint.

7.3 The one-line fix: NPLM on the raw space

Replace NT-Xent-on-the-sphere with the NPLM ratio on the *same raw* z that SIGReg controls, using the critic (14) (or its distance form). Now interaction and marginal share one Euclidean geometry, the gauge argument of §6.3 applies verbatim, and (11)–(13) hold. The upgrade is a single structural swap,

$$\text{normalize} + \text{softmax} \longrightarrow \text{raw} + \text{NPLM}, \quad (26)$$

turning “contrastive features that happen to be Gaussian” into “an embedding that is log-likelihood space.”

7.4 One class, many configurations

All of the above are instances of a single objective

$$\mathcal{L} = \mathcal{L}_{\text{int}}(g; \mathcal{P}) + \lambda \mathcal{L}_{\text{marg}}(z), \quad (27)$$

with four orthogonal knobs:

- **positives** $\mathcal{P}$: instance (augmented views) | supervised (same label) | residual ($z - \bar{z}_y$);
- **critic** g : cosine $\langle \hat{z}, \hat{z}' \rangle / \tau$ | bilinear $\langle z, z' \rangle$ | neg-distance $-\frac{1}{2}\|z - z'\|^2$ ($+b(x) + b(x')$);
- **interaction estimator**: softmax NT-Xent (self-normalized) | NPLM $e^g - 1$ (un-normalized MLE);
- **marginal** $\mathcal{L}_{\text{marg}}$: none | SIGReg $\rightarrow \mathcal{N}(0, I)$ | class-wise SIGReg $\rightarrow \mathcal{N}(m_c, \sigma_c^2 I)$.

The known recipes are corners of this cube:

recipe	positives	critic	estimator	marginal
SimCLR	instance	cosine	softmax	none
SupCon	supervised	cosine	softmax	none
exp-34e hybrid	instance	cosine	softmax	SIGReg
34g supcon+sigreg	supervised	cosine	softmax	SIGReg
LeJEPa-like	pred/target	neg-dist (MSE)	—	SIGReg
NPLM geometry	instance/sup	bilinear/neg-dist	NPLM	SIGReg
class-wise NPLM	supervised	neg-dist	NPLM	class-wise SIGReg

Making `train_simclr_sigreg` a configurable class along these axes means every existing exp-34 arm is a special case and the log-likelihood-space arm is one config flip away — directly comparable under the same probe protocol.

7.5 Practical notes for robustness across configurations

- **Stabilize e^g by clamping, not by subtracting a max.** It is tempting to copy the `sim - sim.max()` guard from `supcon_loss`, but that trick is only valid inside a *log-ratio*: in softmax the subtracted constant cancels, whereas in the NPLM loss there is no enclosing ratio, so subtracting a constant c shifts the minimizer to $g^* + c$ and *destroys the absolute calibration* that is the entire point. Instead clamp the exponent, $e^{\min(g, g_{\max})}$; this tames the blow-up of §5.1 while leaving g^* unchanged in the normal range. (Empirically, with clamping the recovered critic matches the analytic PMI with correlation 0.99 and a mean offset of ≈ 0 ; with max-subtraction the offset runs to hundreds.)
- **Mask same-class pairs from the reference.** With supervised positives, off-diagonal same-class pairs are *positives*, not independent draws; they must be excluded from the $\mathbb{E}_{\text{ref}}[e^g]$ sum or they bias the product-of-marginals estimate — the NPLM analogue of SupCon’s positive masking.
- **Drop the bias under class-wise marginals.** When $\mathcal{L}_{\text{marg}}$ is class-wise SIGReg, the class mean m_c already supplies the per-point offset, so $b(x)$ can be fixed to $-\frac{1}{2}\|z - m_y\|^2$ rather than learned.
- **Fold temperature into g .** A single scale τ (or a learnable scalar on g) subsumes the cosine temperature and the NPLM output scale, keeping the knobs independent and the configurations comparable.